§1-6 Inverse Functions and Logarithms

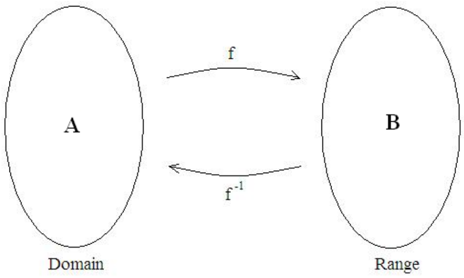
Homework：5,7,13,17,18,20,22,24,28,32,33,34,37,41,51,52,67,68,71

∗ Definition of a function f: Let a correspondence f between 2 sets A and B satisfying the following:

$\forall x \in A \Rightarrow \exists\,!$ (存在唯一) $y \in B$ $s.t.$(使得). $f(x) = y$. Then f is called a function from A to B.

f is 1-1 $\Leftrightarrow$ existence of f^{-1}

f^{-1}代表 f 的反函數， $f^{-1} \neq \dfrac{1}{f}$



Example 1： Let $f(x) = \dfrac{4x-1}{2x+3} = y$. Find the domain and range of f and the inverse of f .

Solution：Domain $= \left\{ x \in R : \; x \neq -\dfrac{3}{2} \right\}$

Let $\dfrac{4x-1}{2x+3} = y \;\; \Rightarrow \;\; x = \dfrac{1+3y}{4-2y} = f^{-1}(y)$

$\Rightarrow$ Range $= \{ y \in R : \;\; y \neq 2 \}$

$\Rightarrow f^{-1}(x) = \dfrac{1+3x}{4-2x}$

Example 2： $m = f(v) = \dfrac{m_0}{\sqrt{1-\dfrac{v^2}{c^2}}}$. Find $f^{-1}(m)$.

Here m_0 is mass at rest, and m is mass with speed v and c is the speed of the light.

Solution：

$$\begin{aligned} m &= \frac{m_0}{\sqrt{1-\dfrac{v^2}{c^2}}} \\ \Rightarrow v &= c\sqrt{1-\frac{m_0^2}{m^2}} \\ &= f^{-1}(m) \end{aligned}$$

Example 3： 解 $e^{5-3x} = 10$.

Solution：

$$5-3x = \ln 10$$

$$\Rightarrow \quad x = \frac{5-\ln 10}{3}$$

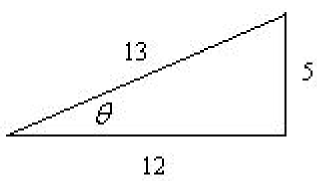
Example 4： $\tan^{-1}\left(\tan\dfrac{4\pi}{3}\right) = \dfrac{\pi}{3}$

Example 5： $\tan^{-1}\tan\sqrt{2}\pi = \left(\sqrt{2}-1\right)\pi$

Example 6： $\sin\left(\tan^{-1}x\right) = \dfrac{x}{\sqrt{1+x^2}}$

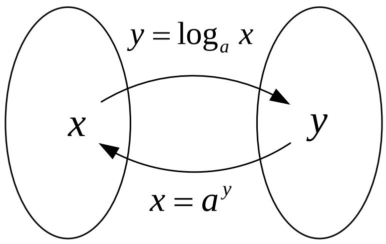
Example 7： $\sin\left(2\sin^{-1}\dfrac{5}{13}\right) = \sin 2\theta = 2\cos\theta\sin\theta = \dfrac{120}{169}$

Solution：



Properties of log.

對數函數：(指數函數的反函數)



(i) $\log_a a^x = x$, $x \in R$; $a^{\log_a x} = x$, $x>0$.

(ii) $\log_a(xy)=\log_a x+\log_a y$, $x,y>0$.

(iii) $\log_a \dfrac{x}{y}=\log_a x-\log_a y$, $x,y>0$

(iv) $\log_a b=\dfrac{\log_c b}{\log_c a}$, $a,b>0$.

(v) $\log_e x=\ln x$